

COMP7503 Multimedia Technologies

Lossy Compression

Dr. Bill LUO

THE UNIVERSITY OF HONG KONG

DEPARTMENT OF COMPUTER SCIENCE

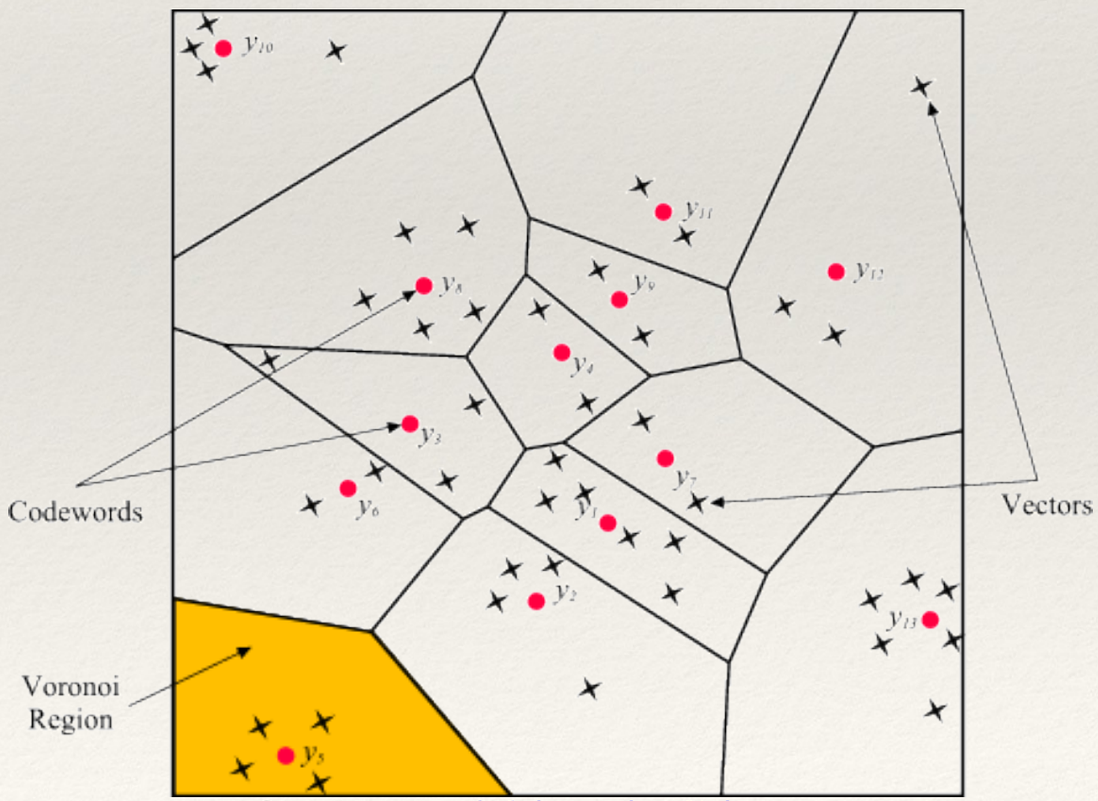
Introduction

- Lossy compression leverages the fact that multimedia data can be gracefully degraded in quality by increasingly losing more information
  - Challenge
    - Tradeoff between quality and cost (cost in terms of storage size or bit-rate)
- Lossy compression achieve compression ratio up to 100:1 (e.g. video compression)
  - Lossless compression usually attains a ratio from 1.3:1 to 2:1



Vector Quantization

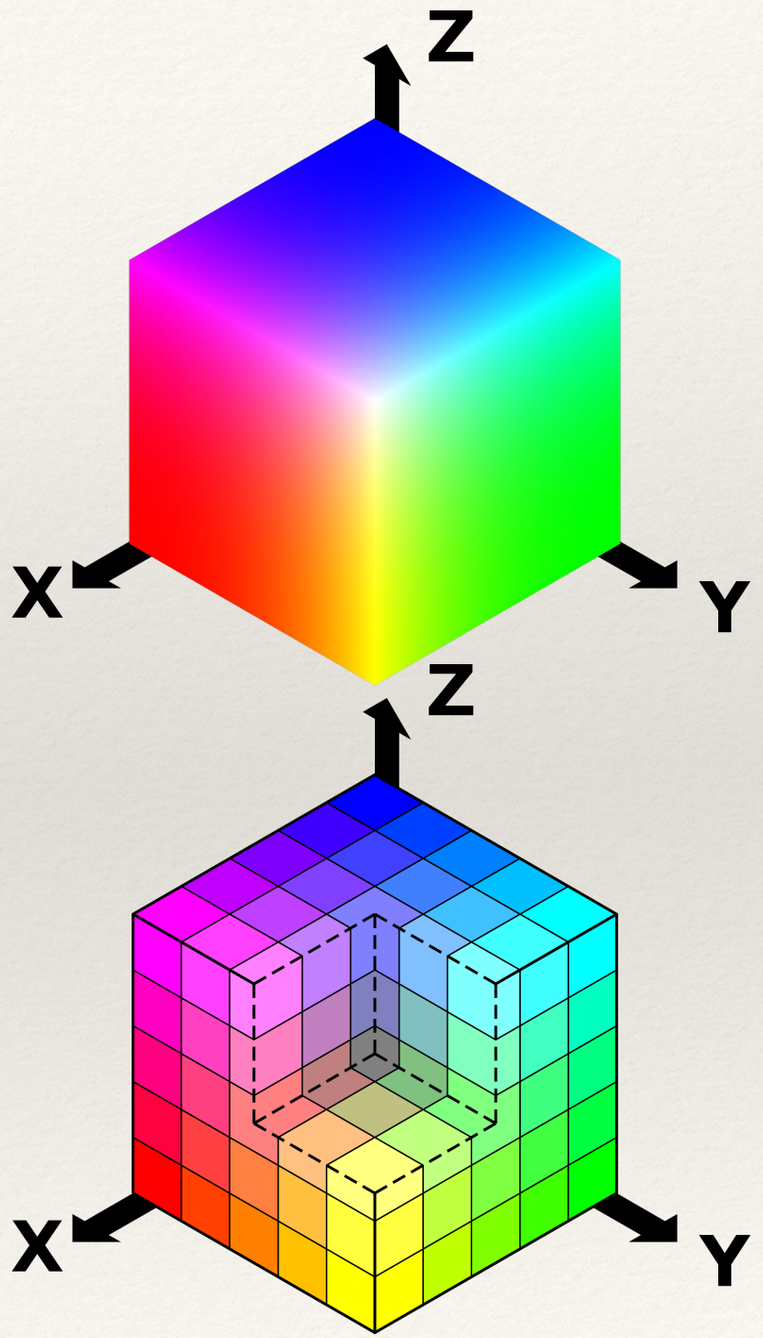
- Vector Quantization is similar to quantisation, but instead of working on a single-valued function, it works on *tuples*, *triples*, or *higher dimensional vectors*
  - Vectors [$\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$] need to be mapped to values [$\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_k$] with $k << n$
- Common approach
  - Linear Quantization
  - K-means
  - X-means



Source: www.cs.ucf.edu/courses/cap5015/vector.ppt

Linear Quantization

- The easiest way to map n vectors [$\mathbf{x}_1$, $\mathbf{x}_2$, …, $\mathbf{x}_n$] to k vectors [$\mathbf{y}_1$, $\mathbf{y}_2$, …, $\mathbf{y}_k$] where $k < n$ is linear quantization
- All of the n vectors distributed evenly into k buckets
  - Usually done by simple arithmetic rounding
- The problem with this method is that
  - n vectors might not be evenly distributed

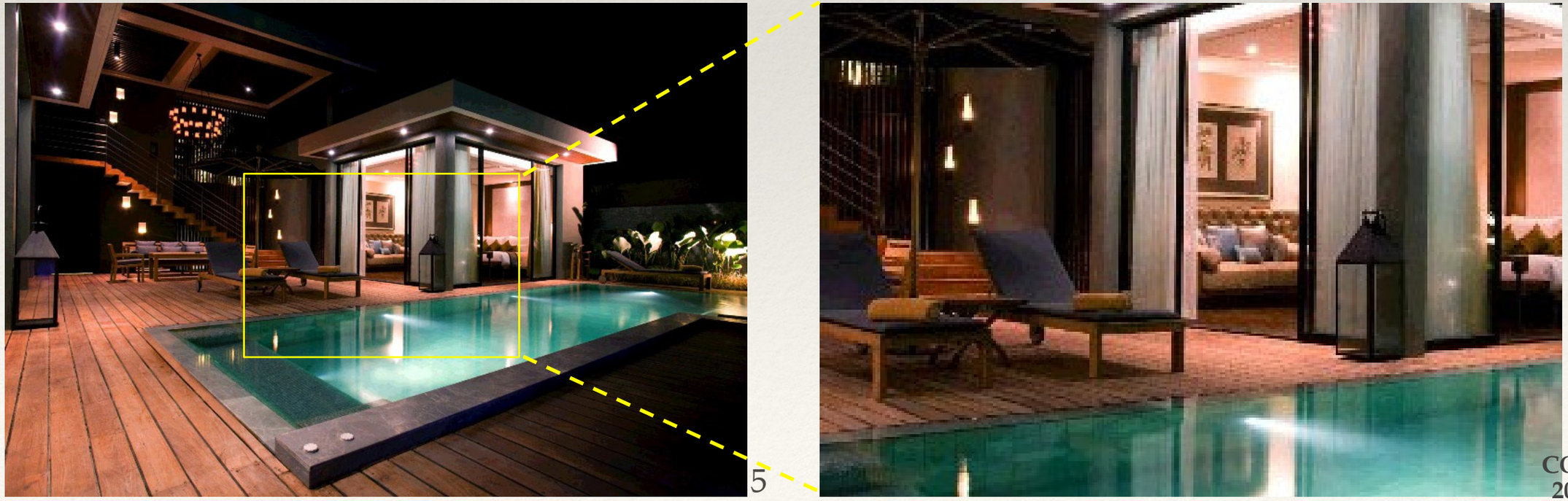


sources://commons.wikimedia.org/wiki/Category:RGB

False Contouring Effect



Color Reduction by Quantisation



False Contouring Effect(Cont'd)



Noise Injection



K-means

- K-means algorithm is well known in statistics and machine learning as a clustering algorithm
- It partition n objects into k clusters, where $k < n$
- The idea of the algorithm is to minimise the intra-cluster variance
  - i.e. the squared distance between each member of a cluster to the mean value of each member of the cluster is to be minimized

$$V = \sum_{i=1}^{k} \sum_{x_j \in S_i} (x_j - \mu_i)^2$$

    - where there are k clusters S_i, $i = 1, 2, \ldots, k$, and μ_i is the centroid or mean point of all points $x_j \in S_i$



Pseudo-code of K-means

// Input: a set of data points **X**, an integer number k

// Output: a set of k data points representing the set μ

k_means(**X**, k)

Choose k initial means μ_i from **X** at random

WHILE (means μ_i are not updated significantly anymore)

// maximization:

FOREACH (sample x_j: from **X**)

assign membership of each element to a mean (closest mean)

// expectation:

FOREACH (mean μ_i)

calculate a new μ_i by averaging x_j that were assigned members

k_means := {μ_i}



Source: https://www.youtube.com/watch?v=BVFG7fd1H30

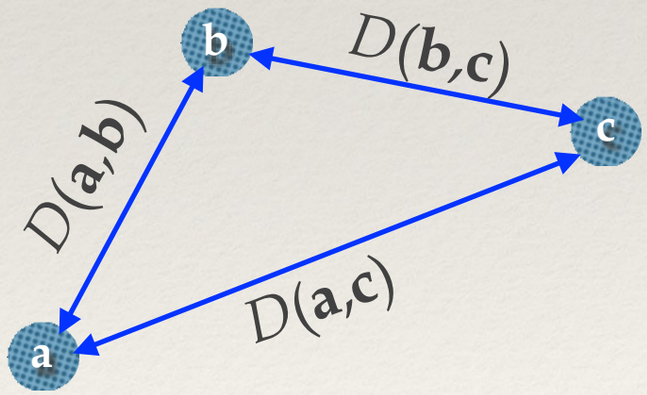
K-means Issues

- Hard to find a proper criteria to judge whether "means μ_i are not updated significantly anymore"
  - Upper bound on number of iterations
  - Change in μ_i measured by some distance measures are smaller than a threshold
  - Computational complexity increases dramatically with large amounts of data
  - Amount of clusters k has to be known in advance
  - How to determine the initial locations of these k clusters



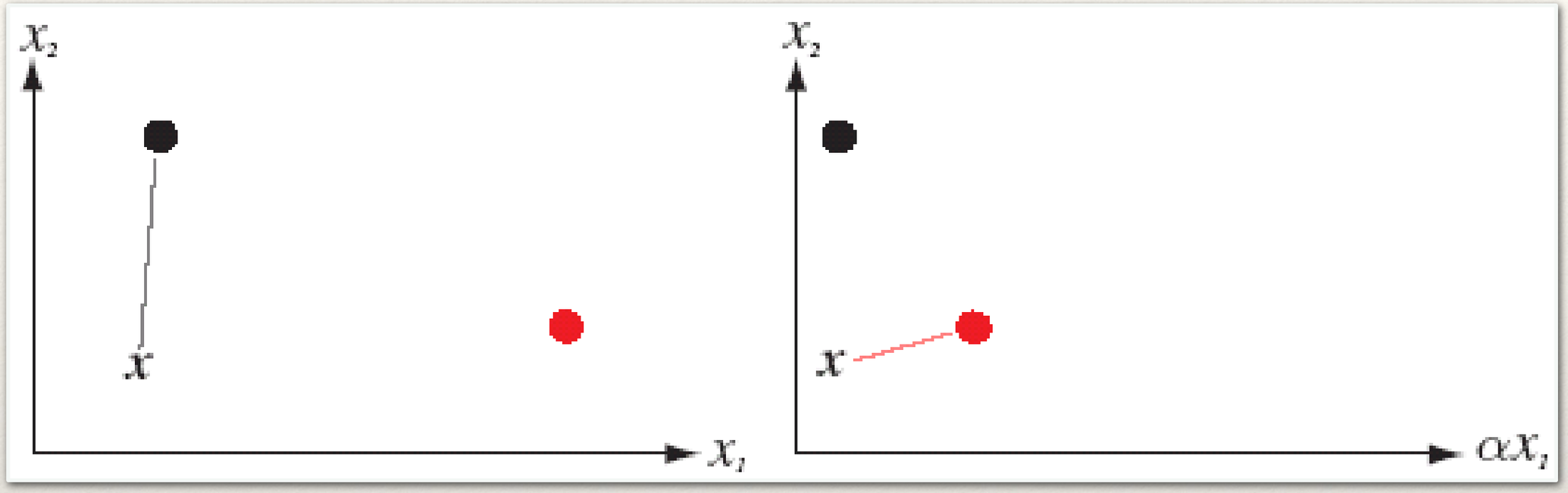
Distance Function

- The K-means algorithm relies on a "distance" function between means and the data points within the cluster
- A metric $D(\cdot, \cdot)$ is a function that gives a generalized scalar distance between two points
  - A metric must have four properties
    - For all vectors **a**, **b** and **c**
      - $D(\mathbf{a}, \mathbf{b}) \geq 0$ (***Nonnegativity***)
      - $D(\mathbf{a}, \mathbf{b}) = 0$ if and only if $\mathbf{a} = \mathbf{b}$ (***Reflexivity***)
      - $D(\mathbf{a}, \mathbf{b}) = D(\mathbf{b}, \mathbf{a})$ (***Symmetry***)
      - $D(\mathbf{a}, \mathbf{b}) + D(\mathbf{b}, \mathbf{c}) \geq D(\mathbf{a}, \mathbf{c})$ (***Triangle inequality***)





Distance Function (Cont'd)



- Euclidean Distance
  - $$d(p, q) = \sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2 + \cdots + (p_i - q_i)^2 + \cdots + (p_n - q_n)^2}.$$
- May not be meaningful when the vector components are differently scaled

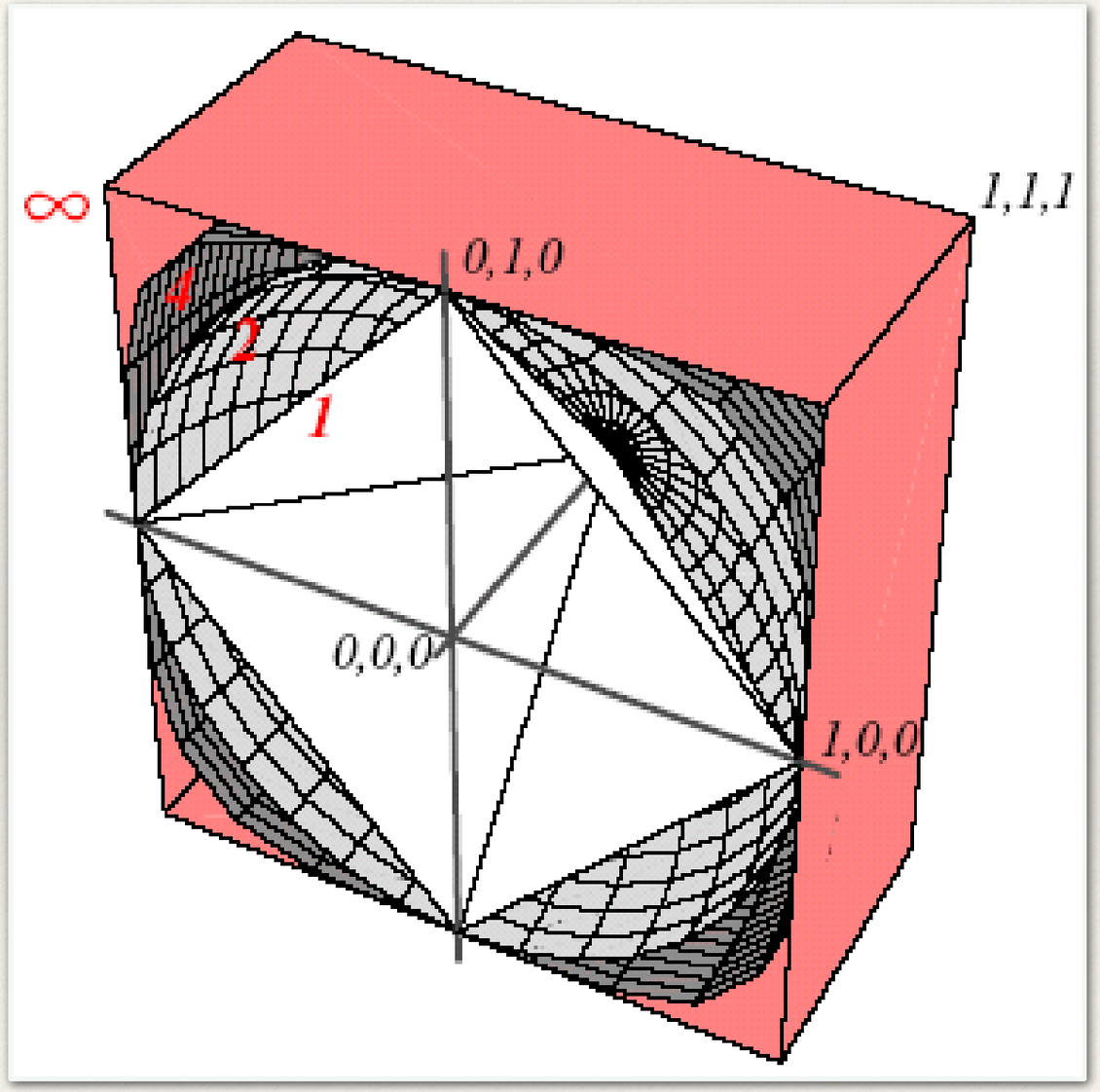


Distance Function (Cont'd)

- Minkowski Metric

$$D(X,Y) = \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{\frac{1}{p}}$$

- L_1 is called the Manhattan or city block distance
- L_2 is the Euclidean distance
- L_∞ corresponds to the maximum of projected distances (onto the axes) between **a** and **b**



X-means

- X-means tries to overcome the problem of the determination of number of clusters k
- The algorithm starts with a k_{min}, which defines the lower bound of the range where k is to be searched, and a k_{max}, which defines the upper bound
  - It tries to exhaustively search for the best $k \in [k_{min}, k_{max}]$ which would yield the best performance according to some scoring mechanism
- For those who are interested, please refer to the paper below
  - https://www.cs.cmu.edu/~dpelleg/download/xmeans.pdf



Pseudo code of X-means

// Input: a set of data points **X**,

// k_{min} and k_{max} denoting lower and upper boundaries for k

// Output: a set of k data points representing the set μ

x_means(**X**)

  $k := k_{min}$

  Run k_means(**X**, k) until it converges

  Score the quality of the clustering

  IF ($k > k_{max}$) stop and report the best scoring clustering during the search

x_means := $\{\mu_i\}_{best}$

- Issue
  - Good scoring mechanism is difficult to determine



Perceptual Quantization

- The idea behind perceptual quantisation is that even though it is highly desirable to reproduce content as accurate as possible, but "accuracy" is not defined based on simple mathematical distance
- Two signals are accurately reproduced if they sound/look the same, even if they have many differences when compared bitwise
- "Accuracy" is measured according to some perceptual model



Sound Pressure Level (SPL)

- Sound Pressure Level (SPL), or L_p, is defined as
  - $$L_p = 10 \log_{10}\left(\frac{p^2}{p^2_{ref}}\right) = 20 \log_{10}\left(\frac{p}{p_{ref}}\right) dB$$
  - Measurement on logarithmic scale using decibels
    - where p is the RMS sound pressure and p_{ref} is a reference sound pressure
    - Typical reference sound pressure for silence is 20 µPa in air and 1 µPa in water
      - p_{ref} is often considered as the threshold of human hearing (roughly the sound of a mosquito flying 3 m away)
- The sound pressure perceived by human ear is non-linear
- Human ear, does not react equally to all frequencies
  - The same sound pressure at a different frequency will be perceived as a different volume level



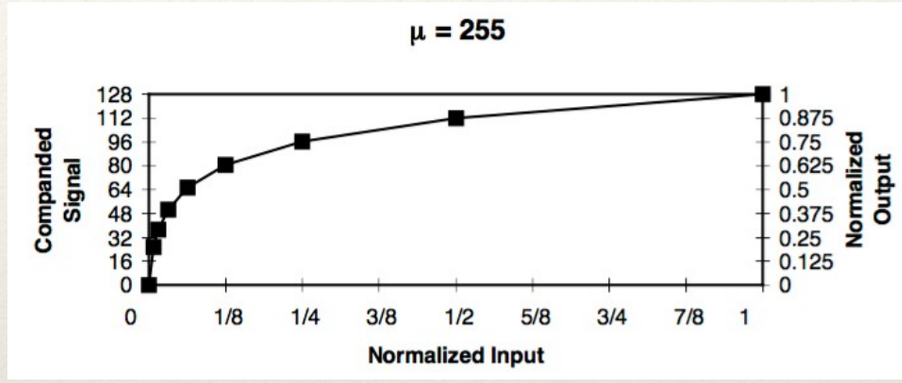
Sound Amplitude Quantization

- The simplest and yet most widely used perceptual quantization techniques for audio data include the A-law and μ-law algorithms
- According to Weber-Fechner law, sound intensity is perceived logarithmically
  - A sound must be twice as intense to be perceived as a constant factor more intense in the human ear
- The idea behind the A-law and μ-law algorithms is to exactly exploit this fact:
  - Rather than quantizing the intensity of the sounds, that is, the sample values, linearly, they are quantized logarithmically

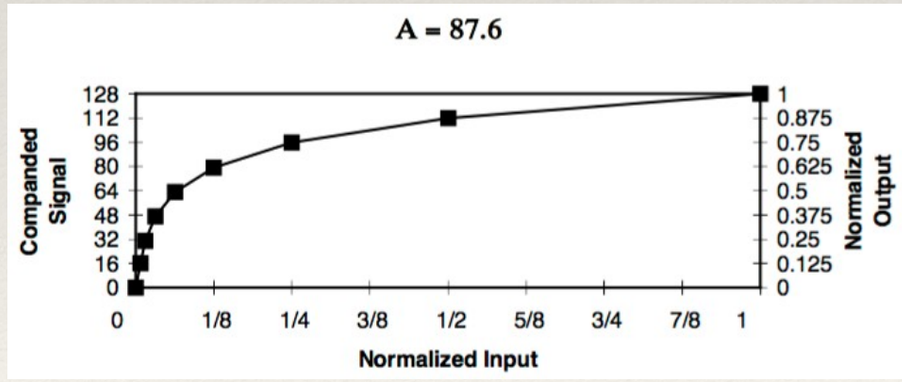


μ-law and A-law

- Companding (Compression and Expanding) formula for μ-law
  - $$F(x) = sgn(x)\frac{\ln(1+\mu|x|)}{\ln(1+\mu)} \qquad -1 \le x \le 1$$
  - where μ is usually 255 for eight bits and sgn is the sign function.



- Companding formula for A-law
  - $$F(x) = \mathrm{sgn}(x)\begin{cases} \dfrac{A|x|}{1+\ln(A)} & |x| < \dfrac{1}{A} \\ \dfrac{1+\ln(A|x|)}{1+\ln(A)} & \dfrac{1}{A} \le |x| \le 1 \end{cases}$$
  - where A=87.6



μ-law and A-law (Cont'd)

- Both are conceptually similar
  - μ-law provides slightly better compression
  - A-law provides slightly better quality
- Telephone companies prefer A-law
- μ-law become a quasi standard for low-bandwidth voice recordings, as it is the default encoding of Sun's audio file format (file extension, ".au")
  - It has been standardised as ITU-T Recommendation G.711



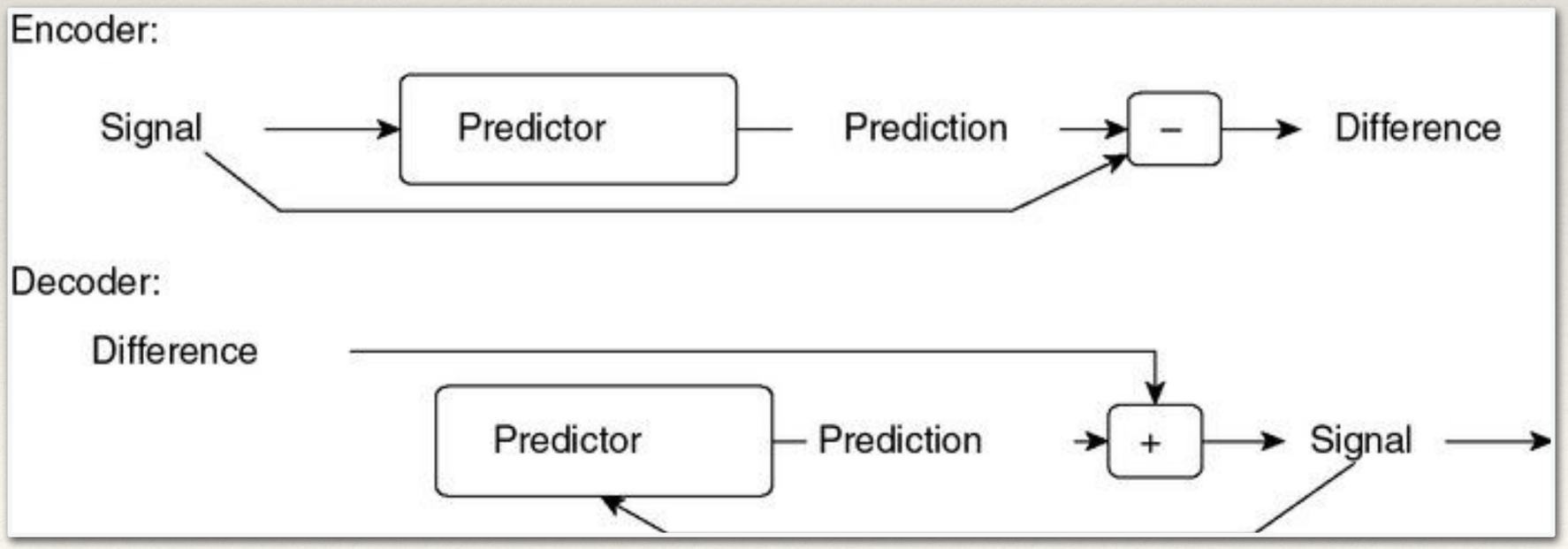
Visual and Motion Quantization

- Like ears, different quantities sensed by eyes scale logarithmically with intensity
  - That means using a logarithmic scale for brightness, similar to μ-law, could be effectively used for image data as well
- A video is a quantised version of the reality it represents
  - The human eyes can fuse images into moving scenes if the images are presented with about one-thirtieth of a second between them
    - That corresponds to a frame rate of about 30 Hz
  - The coexistence of a lip-synchronous audio track generally allows for lower frame rates because human brain is good at filling missing information across modalities
  - Bringing down the frame rate is an effective way to reduce video storage size



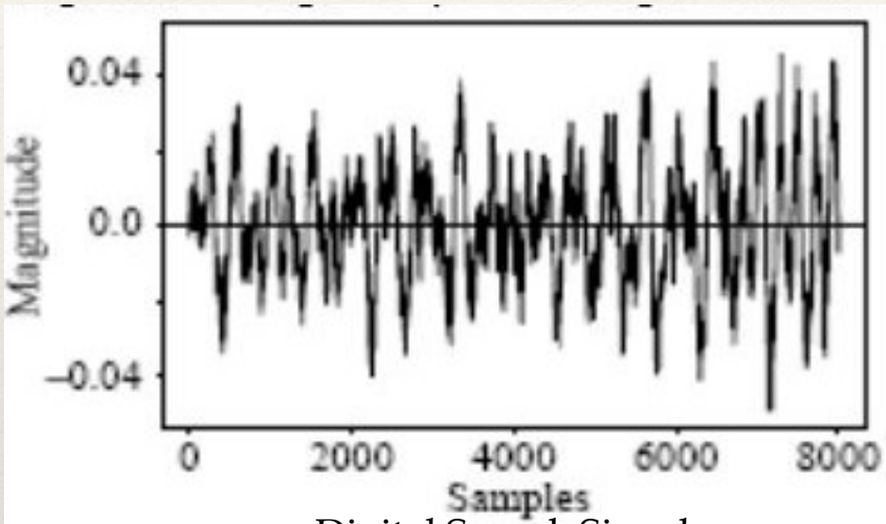
Differential Coding

- Differential Coding leverages global knowledge about the properties of the signal to encode
- It is widely used for audio, image and video compression
- It can scales very well from lossless to entirely lossy

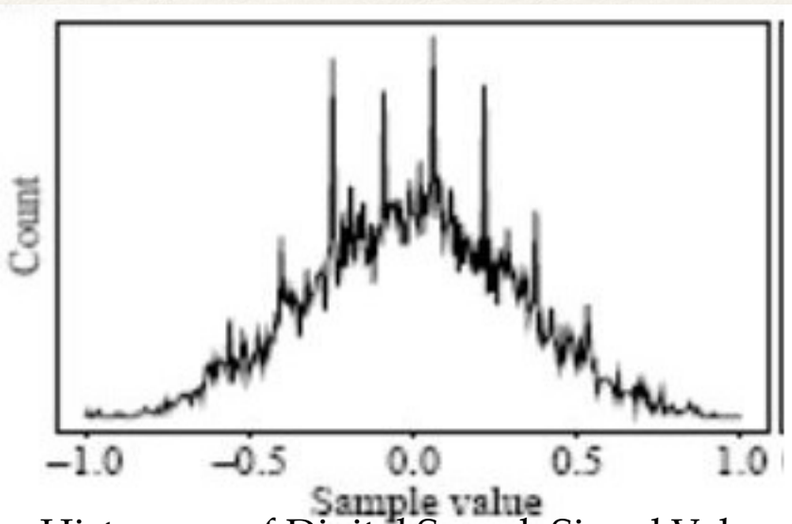


Differential Coding (Cont'd)

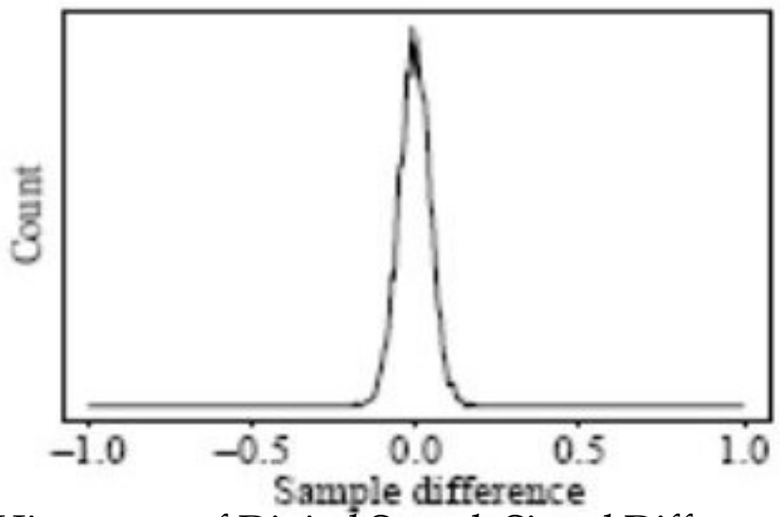
- The simplest predictor: Predict the next sample to be equal to the current sample
- Differencing concentrates the histogram



Digital Speech Signal



Histogram of Digital Speech Signal Values



Histogram of Digital Speech Signal Differences

Source: http://slideplayer.com/slide/5020324/

Differential Coding (Cont'd)

- The main component of a differential encoder is the predictor
  - The predictor takes *n* input samples of the signal to predict the next *m* samples
  - A good encoder will produce very small differences that do not need many bits for encoding
- Lossy differential coding can be achieved by quantising the difference values
  - This achieves higher compression ratio
- Difference encoding is usually combined with entropy encoding for better compression performance



Adaptive Differential Pulse Code Modulation (ADPCM)

- ADPCM is a widespread audio format used mainly for speech encoding, which has been standardized in ITU-T G.726
- The idea behind the difference encoding of audio is that sound other than noise follows a rather predictable wave pattern
- Still, in some cases, sample values might drop from very high amplitude to very low amplitude
- ADPCM is adaptive in the sense that, it tries to predict the amplitude level in the next couple of samples

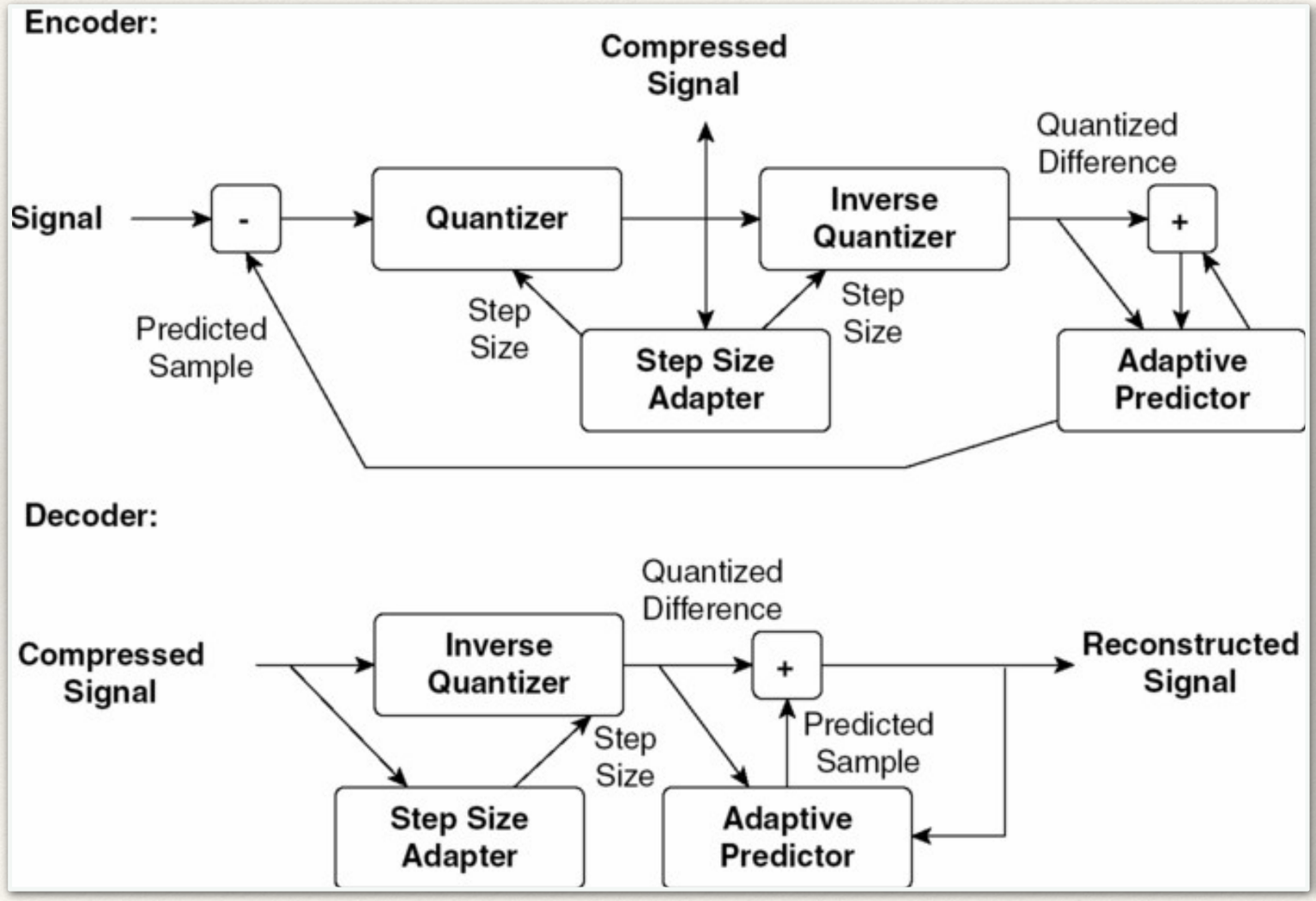


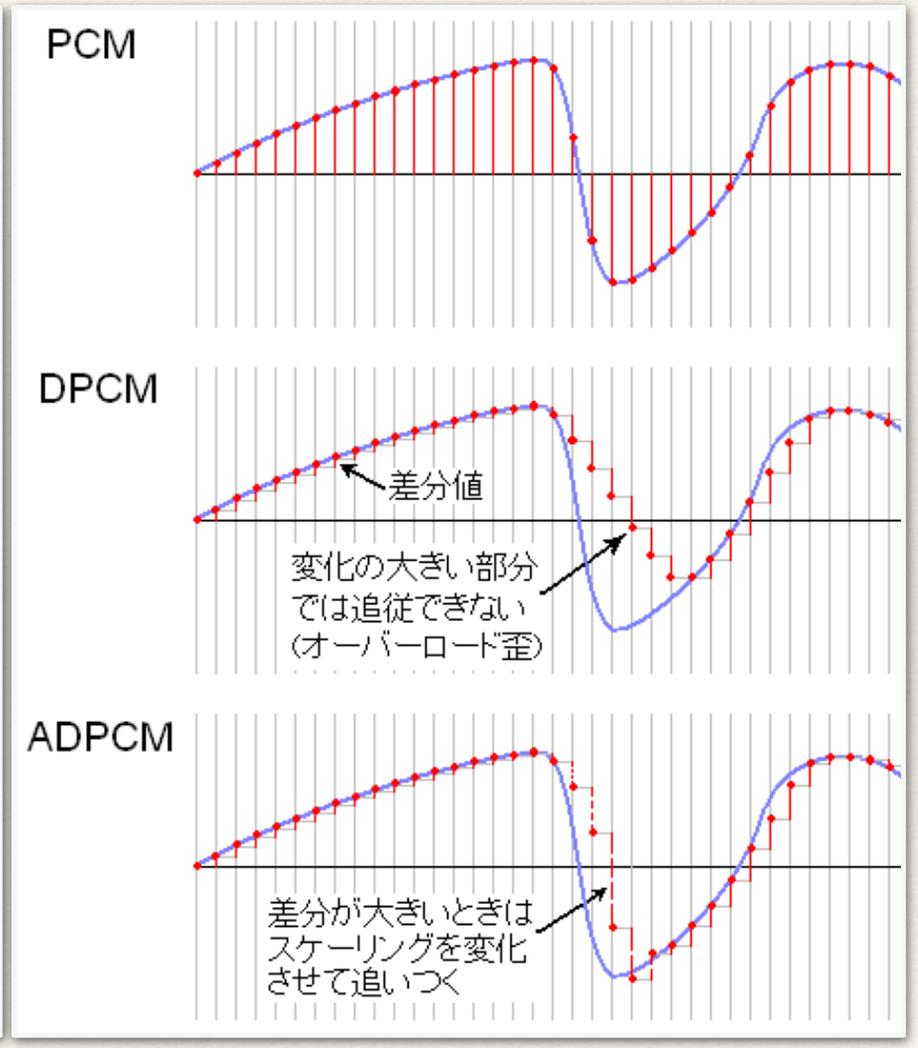
ADPCM (Cont'd)

- The ADPCM algorithm is used to map a series of twelve-bit μ-law (or A-law) PCM samples into a series of four-bit ADPCM samples
  - One bit for the sign, and three bits for the difference
- Given the original sample values as input, ADPCM predicts both the next sample and the step size
  - A large step size means that the audio sample differences are large; a small step size means the differences are small
- ADPCM achieves a decent compression rate ~4:1, and is very usable for speech signals



ADPCM (Cont'd)

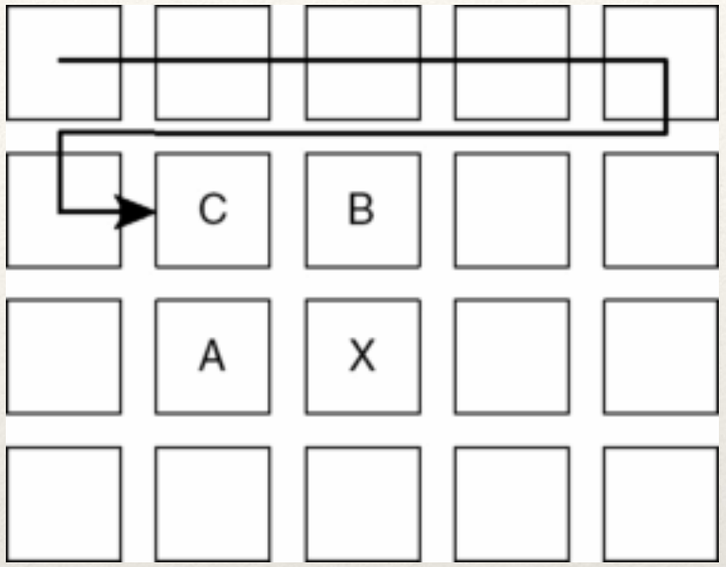




Source: http://elm-chan.org/junk/adpcm/report.html

Differential Coding in Images

- PNG image format uses a differential encoding step before an LZ-derivate entropy encoder is used
- The algorithm predicts the color of each pixel based on the colours of previous neighbouring pixels
  - In the hope that the differences from prediction are highly biased towards zero
- Five possible states for prediction
  - State is specified on a per scan-line basis



State	Name	Predicted Value
0	None	Zero (pass through)
1	Sub	Value of A
2	Up	Value of B
3	Average	Rounded Down of Mean of A and B
4	Paeth	A, B, or C, whichever is closest to p = a + b - c